

**PROJECT REPORT for B. TECH. FINAL
YEAR MAJOR PROJECT**

titled

***Intelligent Email & SMS Spam Detection
using Machine Learning***

to be presented at

GRAPHIC ERA HILL UNIVERSITY, DEHRADUN

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6.1 Dataset Collection and Cleaning – 100% Completed

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1. Introduction

Today, Spam has become a major problem in communication over the internet. It has been accounted that around 55% of all emails are reported as spam and the number has been growing steadily. Spam which is also known as unsolicited bulk email has led to the increasing use of email as email provides the perfect ways to send the unwanted advertisement or junk newsgroup posting at no cost for the sender. This chance has been extensively exploited by irresponsible organizations and resulting to clutter the mail boxes of millions of people all around the world.

Spam has been a major concern given the offensive content of messages, spam is a waste of time. End user is at risk of deleting legitimate mail by mistake. Moreover, spam also impacted the economy which led some countries to adopt legislation.

Text classification is used to determine the path of incoming mail/message either into inbox or straight to spam folder. It is the process of assigning categories to text according to its content. It is used to organized, structures and categorize text. It can be done either manually or automatically. Machine learning automatically classifies the text in a much faster way than manual technique. Machine learning uses pre-labelled text to learn the different associations between pieces of text and its output. It used feature extraction to transform each text to numerical representation in form of vector which represents the frequency of word in predefined dictionary.

Text classification is important to structure the unstructured and messy nature of text such as documents and spam messages in a cost-effective way. Machine learning can make more accurate precisions in real-time and help to improve the manual slow process to much better and faster analysing big data. It is important especially to a company to analyse text data, help inform business decisions and even automate business processes.

In this project, machine learning techniques are used to detect the spam message of a mail. Machine learning is where computers can learn to do something

without the need to explicitly program them for the task.

It uses data and produce a program to perform a task such as classification. Compared to knowledge engineering, machine learning techniques require messages that have been successfully pre-classified. The pre-classified messages make the training dataset which will be used to fit the learning algorithm to the model in machine learning studio.

A combination of algorithms are used to learn the classification rules from messages. These algorithms are used for classification of objects of different classes. These algorithms are provided with pre labelled data and an unknown text. After learning from the prelabelled data each of these algorithms predict which class the unknown text may belong to and the category predicted by majority is considered as final.

2. Objectives and Scope

2.1 Problem Statement

Spammers are in continuous war with Email service providers. Email service providers implement various spam filtering methods to retain their users, and spammers are continuously changing patterns, using various embedding tricks to get through filtering. These filters can never be too aggressive because a slight misclassification may lead to important information loss for consumer. A rigid filtering method with additional reinforcements is needed to tackle this problem.

2.2 Objectives

The objectives of this project are

- To create an ensemble algorithm for classification of spam with highest possible accuracy.
- To study on how to use machine learning for spam detection.
- To study how natural language processing techniques can be implemented in spam detection.
- To provide user with insights of the given text leveraging the created algorithm and NLP.

2.3 Project Scope

This project needs a coordinated scope of work.

- i. Combine existing machine learning algorithms to form a better ensemble algorithm.
- ii. Clean, process and make use of the dataset for training and testing the model created.
- iii. Analyse the texts and extract entities for presentation.

2.4 Limitations

This Project has certain limitations.

- i. This can only predict and classify spam but not block it.
- ii. Analysis can be tricky for some alphanumeric messages and it may struggle with entity detection.
- iii. Since the data is reasonably large it may take a few seconds to classify and analyse the message.

3. Experimentation and Methods

3.1 Introduction

This chapter will explain the specific details on the methodology being used to develop this project. Methodology is an important role as a guide for this project to make sure it is in the right path and working as well as plan. There is different type of methodology used in order to do spam detection and filtering. So, it is important to choose the right and suitable methodology thus it is necessary to understand the application functionality itself.

3.2 System Architecture

The application overview has been presented below and it gives a basic structure of the application.

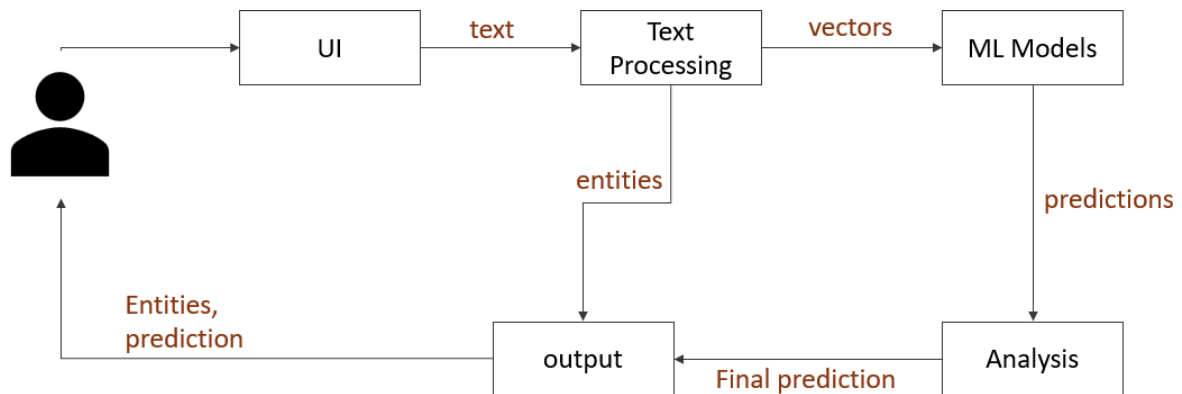


fig no. 4.1 Architecture

The UI, Text processing and ML Models are the three important modules of this project. Each Module's explanation has been given in the later sections of this chapter.

A more complicated and detailed view of architecture is presented in the workflow section.

3.3 Modules and Explanation

The Application consists of three modules.

- i. UI
- ii. Machine Learning
- iii. Data Processing

I. UI Module

- a. This Module contains all the functions related to UI(user interface).
- b. The user interface of this application is designed using Streamlit library from python based packages.
- c. The user inputs are acquired using the functions of this library and forwarded to data processing module for processing and conversion.
- d. Finally the output from ML module is sent to this module and from this module to user in visual form.

II. Machine Learning Module

- a. This module is the main module of all three modules.
- b. This modules performs everything related to machine learning and results analysis.
- c. Some main functions of this module are
 - Training machine learning models.
 - Testing the model
 - Determining the respective parameter values for each model.
 - Key-word extraction.
 - Final output calculation
- d. The output from this module is forwarded to UI for providing visual response to user

III. Data Processing Module

- a. The raw data undergoes several modifications in this module for further process.
- b. Some of the main functions of this module includes
 - Data cleaning
 - Data merging of datasets
 - Text Processing using NLP
 - Conversion of text data into numerical data(feature vectors).
 - Splitting of data.
- c. All the data processing is done using Pandas and NumPy libraries.
- d. Text processing and text conversion is done using NLTK and scikit-learn libraries.

3.4 Requirements

Hardware Requirements

PC/Laptop

Ram – 8 Gig

Storage – 100-200 Mb

Software Requirements

OS – Windows 7 and above

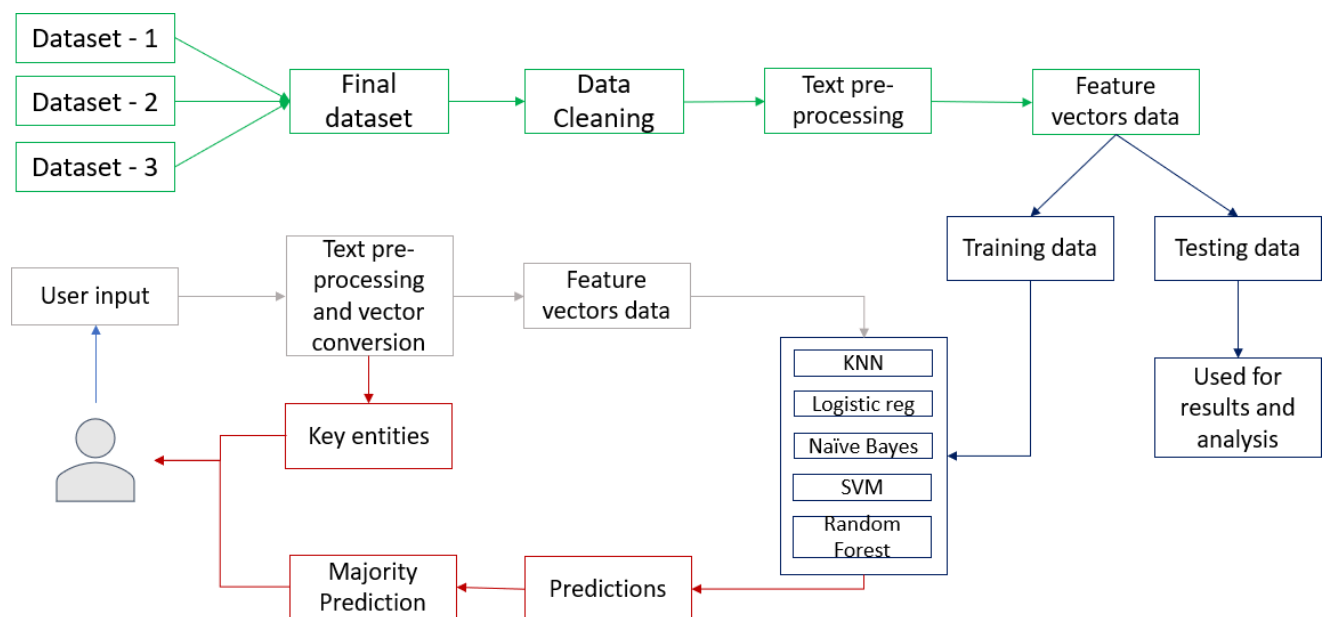
Code Editor – Pycharm, VS Code, Built in IDE

Anaconda environment with packages nltk, numpy, pandas, sklearn, tkinter, nltk data.

Supported browser such as chrome, firefox, opera etc..

3.5 WorkFlow

fig no. 4.2 Workflow



In the above architecture, the objects depicted in Green belong to a module called Data Processing. It includes several functions related to data processing, natural Language Processing. The objects depicted in Blue belong to the Machine Learning module. It is where everything related to ML is embedded. The red objects represent final results and outputs.

Data Collection and Description

- Data plays an important role when it comes to prediction and classification, the more the data the more the accuracy will be.
- The data used in this project is completely open-source and has been taken from various resources like Kaggle and UCI
- For the purpose of accuracy and diversity in data multiple datasets are taken. 2 datasets containing approximately over 12000 mails and their labels are used for training and testing the application.
- 6000 spam mails are taken for generalisation of data and to increase the accuracy.

Data Description

Dataset : enronSpamSubset.

Source : Kaggle

Description : this dataset is part of a larger dataset called enron. This dataset contains a set of spam and non-spam emails with 0 for non spam and 1 for spam in label attribute.

Composition :

Unique values : 9687

Spam values : 5000

Non-spam values :

4687

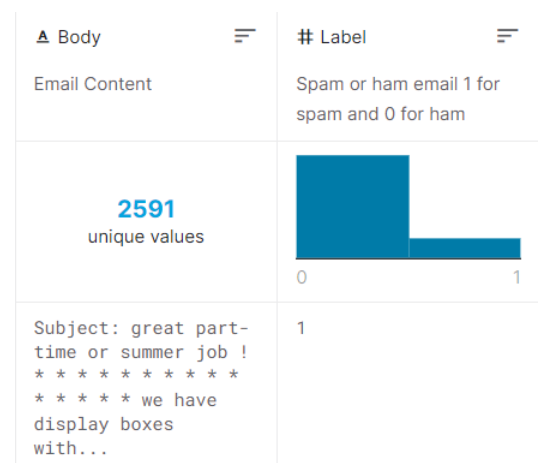


fig no. 4.3 enron spam

Dataset : lingspam.

Source : Kaggle

Description : This dataset is part of a larger dataset called Enron1 which contains emails classified as spam or ham(not-spam).

Composition :

Unique values : 2591

Spam values : 419

Non-spam values : 2172

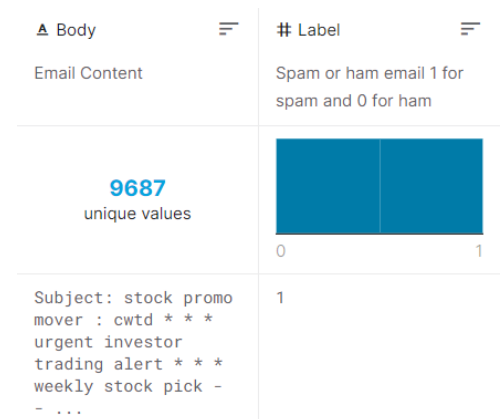


fig no. 4.4 lingspam

3.6 Data Processing

Overall data processing

It consists of two main tasks

Dataset cleaning

It includes tasks such as removal of outliers, null value removal, removal of unwanted features from data.

Dataset Merging

After data cleaning, the datasets are merged to form a single dataset containing only two features(text, label).

Data cleaning, Data Merging these procedures are completely done using Pandas library.

Textual data processing

- **Tag removal**

Removing all kinds of tags and unknown characters from text using regular expressions through Regex library.

- **Sentencing, tokenization**

Breaking down the text(email/SMS) into sentences and then into tokens(words). This process is done using NLTK pre-processing library of python.

- **Stop word removal**

Stop words such as of , a ,be , ... are removed using stopwords NLTK library of python.

- **Lemmatization**

Words are converted into their base forms using lemmatization and pos-tagging

This process gives key-words through entity extraction.

This process is done using chunking in regex and NLTK lemmatization.

- **Sentence formation**

The lemmatized tokens are combined to form a sentence.

This sentence is essentially a sentence converted into its base form and removing stop words.

Then all the sentences are combined to form a text.

While the overall data processing is done only to datasets, the textual processing is done to both training data, testing data and also user input data.

Feature Vector Formation

- The texts are converted into feature vectors(numerical data) using the words present in all the texts combined
- This process is done using countvectorization of **NLTK** library.
- The feature vectors can be formed using two language models Bag of Words and Term Frequency-inverse Document Frequency.

Bag of Words

Bag of words is a language model used mainly in text classification. A bag of words represents the text in a numerical form.

The two things required for Bag of Words are

- A vocabulary of words known to us.
- A way to measure the presence of words.

Ex: a few lines from the book “A Tale of Two Cities” by Charles Dickens.

**“It was the best of times,
It was the worst of times,
It was the age of wisdom,
It was the age of
foolishness”**

The unique words here (ignoring case and punctuation) are:

[“it”, “was”, “the”, “best”, “of”, “times”, “worst”, “age”, “wisdom”, “foolishness”]

The next step is scoring words present in every document.

After scoring the four lines from the above stanza can be represented in vector form as

“It was the best of times” = [1, 1, 1, 1, 1, 1, 0, 0, 0, 0]

“it was the worst of times” = [1, 1, 1, 0, 1, 1, 1, 0, 0, 0]

“it was the age of wisdom” = [1, 1, 1, 0, 1, 0, 0, 1, 1, 0]

“it was the age of foolishness” = [1, 1, 1, 0, 1, 0, 0, 1, 0, 1]

This is the main process behind the bag of words but in reality the vocabulary even from a couple of documents is very large and words repeating frequently and important in nature are taken and remaining are removed during the text processing stage.

Term Frequency-inverse document frequency

Term frequency-inverse document frequency of a word is a measurement of the importance of a word. It compares the repentence of words to the collection of documents and calculates the score. Terminology for the below formulae:

t – term(word)

d – document(set of words)

N – count of documents

The TF-IDF process consists of various activities listed below.

i) Term Frequency

The count of appearance of a particular word in a document is called term frequency

$$tf(t, d) = \text{count of } t \text{ in } d / \text{number of words in } d$$

ii) Document Frequency

Document frequency is the count of documents the word was detected in. We consider one instance of a word and it doesn't matter if the word is present multiple times.

$$df(t) = \text{occurrence of } t \text{ in documents}$$

iii) Inverse Document Frequency

- IDF is the inverse of document frequency.
- It measures the importance of a term t considering the information it contributes. Every term is considered equally important but certain terms such as (are, if, a, be, that, ..) provide little information about the document. The inverse document frequency factor reduces the importance of words/terms that has higher recurrence and increases the importance of words/terms that are rare.

$$idf(t) = N/df$$

Finally, the TF-IDF can be calculated by combining the term frequency and inverse document frequency.

$$tf_idf(t, d) = tf(t, d) * \log(N/(df + 1))$$

the process can be explained using the following example:

“Document 1 It is going to rain today.

Document 2 Today I am not going outside.

Document 3 I am going to watch the season premiere.”

The Bag of words of the above sentences is

[going:3, to:2, today:2, i:2, am:2, it:1, is:1, rain:1]

Then finding the term frequency

table no. 4.1 Term frequency

Words	IDF Value
Going	$\log(3/3)$
To	$\log(3/2)$
Today	$\log(3/2)$
I	$\log(3/2)$
Am	$\log(3/2)$
It	$\log(3/1)$
Is	$\log(3/1)$
rain	$\log(3/1)$

Then finding the inverse document frequency

table no. 4.2 inverse document frequency

Words	Document1	Document2	Document3
Going	0.16	0.16	0.12
To	0.16	0	0.12
Today	0.16	0.16	0
I	0	0.16	0.12
Am	0	0.16	0.12
It	0.16	0	0
Is	0.16	0	0
rain	0.16	0	0

Applying the final equation the values of tf-idf becomes

Words/ documents	going	to	Today	i	am	if	it	rain
Document1	0	0.07	0.07	0	0	0.17	0.17	0.17
Document2	0	0	0.07	0.07	0.07	0	0	0
Document3	0	0.05	0	0.05	0.05	0	0	0

table no. 4.3 TF-IDF

Using the above two language models the complete data has been converted into two kinds of vectors and stored into a csv type file for easy access and minimal processing.

Data Splitting

The data splitting is done to create two kinds of data Training data and testing data.

Training data is used to train the machine learning models and testing data is used to test the models and analyse results. 80% of total data is selected as testing data and remaining data is testing data.

4. Machine Learning

4.1 Introduction

Machine Learning is process in which the computer performs certain tasks without giving instructions. In this case the models takes the training data and train on them.

Then depending on the trained data any new unknown data will be processed based on the ruled derived from the trained data.

After completing the countvectorization and TF-IDF stages in the workflow the data is converted into vector form(numerical form) which is used for training and testing models.

For our study various machine learning models are compared to determine which method is more suitable for this task. The models used for the study include Logistic Regression, Naïve Bayes, Random Forest Classifier, K Nearest Neighbors, and Support Vector Machine Classifier and a proposed model which was created using an ensemble approach.

4.2 Algorithms

a combination of 5 algorithms are used for the classifications.

A: Naïve Bayes Classifier

A naïve Bayes classifier is a supervised probabilistic machine learning model that is used for classification tasks. The main principle behind this model is the Bayes theorem.

Bayes Theorem:

Naive Bayes is a classification technique that is based on Bayes' Theorem with an assumption that all the features that predict the target value are independent of each other. It calculates the probability of each class and then picks the one with the highest probability.

Naive Bayes classifier assumes that the features we use to predict the target are independent and do not affect each other. Though the independence assumption is never correct in real-world data, but often works well in practice. so that it is called “Naive” [14].

$$P(A|B) = (P(B|A)P(A))/P(B)$$

$P(A|B)$ is the probability of hypothesis A given the data B. This is called the posterior probability.

$P(B|A)$ is the probability of data B given that hypothesis A was true.

$P(A)$ is the probability of hypothesis A being true (regardless of the data). This is called the prior probability of A.

$P(B)$ is the probability of the data (regardless of the hypothesis) [15].

Naïve Bayes classifiers are mostly used for text classification. The limitation of the Naïve Bayes model is that it treats every word in a text as independent and is equal in importance but every word cannot be treated equally important because articles and nouns are not the same when it comes to language. But due to its classification efficiency, this model is used in combination with other language processing techniques.

B: Random Forest Classifier

Random Forest classifier is a supervised ensemble algorithm. A random forest consists of multiple random decision trees. Two types of randomnesses are built into the trees. First, each tree is built on a random sample from the original data. Second, at each tree node, a subset of features is randomly selected to generate the best split [16].

Decision Tree:

The decision tree is a classification algorithm based completely on features. The tree repeatedly splits the data on a feature with the best information gain. This process continues until the information gained remains constant. Then the unknown data is evaluated feature by feature until categorized. Tree pruning techniques are used for improving accuracy and reducing the overfitting of data.

Several decision trees are created on subsets of data the result that was given by the majority of trees is considered as the final result. The number of trees to be created is determined based on accuracy and other metrics through iterative methods. Random forest classifiers are mainly used on condition-based data but it works for text if the text converted into numerical form.

C: Logistic Regression

Logistic Regression is a “Supervised machine learning” algorithm that can be used to model the probability of a certain class or event. It is used when the data is linearly separable and the outcome is binary or dichotomous [17]. The probabilities are calculated using a sigmoid function.

For example, let us take a problem where data has n features.

We need to fit a line for the given data and this line can be represented by the equation

$$z = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + \dots + b_n x_n$$

here $z = \text{odds}$

generally, odds are calculated as

$$\text{odds} = p(\text{event occurring}) / p(\text{event not occurring})$$

Sigmoid Function:

A sigmoid function is a special form of logistic function hence the name logistic regression.

The logarithm of odds is calculated and fed into the sigmoid function to get continuous probability ranging from 0 to 1.

The logarithm of odds can be calculated by

$$\log(\text{odds}) = \text{dot}(\text{features}, \text{coefficients}) + \text{intercept}$$

and these $\log_odds$ are used in the sigmoid function to get probability.

$$h(z) = 1 / (1 + e^{(-z)})$$

The output of the sigmoid function is an integer in the range 0 to 1 which is used to determine which class the sample belongs to. Generally, 0.5 is considered as the limit below which it is considered a NO, and 0.5 or higher will be considered a YES. But the border can be adjusted based on the requirement.

D: K-Nearest Neighbors

KNN is a classification algorithm. It comes under supervised algorithms. All the data points are assumed to be in an n-dimensional space. And then based on neighbors the category of current data is determined based on the majority.

Euclidian distance is used to determine the distance between points.
The distance between 2 points is calculated as

$$d = \sqrt{[(x_2 - x_1)]^2 + [(y_2 - y_1)]^2}$$

The distances between the unknown point and all the others are calculated. Depending on the K provided k closest neighbors are determined. The category to which the majority of the neighbors belong is selected as the unknown data category.

If the data contains up to 3 features then the plot can be visualized. It is fairly slow compared to other distance-based algorithms such as SVM as it needs to determine the distance to all points to get the closest neighbors to the given point.

E: Support Vector Machines(SVM)

It is a machine learning algorithm for classification. Decision boundaries are drawn between various categories and based on which side the point falls to the boundary the category is determined.

Support Vectors:

The vectors closer to boundaries are called support vectors/planes. If there are n categories then there will be n+1 support vectors. Instead of points, these are called vectors because they are assumed to be starting from the origin. The distance between the support vectors is called margin. We want our margin to be as wide as possible because it yields better results.

There are three types of boundaries used by SVM to create boundaries.

Linear: used if the data is linearly separable.

Poly: used if data is not separable. It creates any data into 3-dimensional data.

Radial: this is the default kernel used in SVM. It converts any data into infinite-dimensional data.

5. CONCLUSION

Email spam has become one of the most significant challenges in modern digital communication systems due to the rapid expansion of internet usage and the widespread reliance on email services. The increasing volume of unsolicited and malicious emails has resulted in serious security risks, reduced productivity, and inefficient utilization of network and system resources. Traditional rule-based spam filtering methods are no longer effective in this dynamic environment, as they fail to adapt to continuously evolving spam strategies employed by attackers.

This project presents an intelligent Artificial Intelligence–driven spam detection system that leverages Machine Learning techniques to automatically classify emails as spam or legitimate (ham). Unlike conventional approaches that depend on static rules, the proposed system learns directly from historical email data and identifies hidden patterns and correlations within the data. This learning-based approach enables the system to adapt to new spam trends, making it more scalable, flexible, and reliable.

The system employs machine learning pipelines that transform raw email content into structured numerical representations, enabling efficient model training and prediction. Multiple supervised machine learning algorithms—including Naïve Bayes, Logistic Regression, Random Forest, K-Nearest Neighbors, and Support Vector Machines—were implemented and trained on the same dataset to ensure consistency and fairness in evaluation. This comparative approach helped identify the strengths and weaknesses of individual classifiers in the context of spam detection.

Experimental analysis revealed that TF-IDF-based feature representation significantly enhances the performance of machine learning models compared to traditional frequency-based representations. Moreover, the ensemble learning strategy, which aggregates predictions from multiple machine learning models using majority voting, achieved superior accuracy and robustness compared to standalone classifiers. The ensemble model demonstrated high accuracy, precision, and F1-score, validating its effectiveness as an AI-powered spam detection solution.

In conclusion, this project successfully demonstrates that Machine Learning and Artificial Intelligence provide a powerful and adaptive solution to email spam detection. By combining multiple learning algorithms and optimized feature representation techniques, the proposed system minimizes false positives, improves detection accuracy, and adapts to evolving spam patterns. The developed system significantly enhances email security and user experience, proving the practical value of AI-driven decision-making in real-world communication systems.

6. Current Status of the Project

The development of the Spam Detection in Emails using Artificial Intelligence and Machine Learning project has progressed through several important stages. Each stage contributes to the overall functionality, accuracy, and reliability of the system. The current progress of the project is elaborated below:

1. Dataset Collection and Cleaning – 100% Completed

The dataset required for training and testing the spam detection models has been successfully collected from reliable and publicly available sources. The dataset contains a balanced set of spam and legitimate (ham) emails, ensuring effective learning by machine learning algorithms.

- Data cleaning has been fully completed, which involved:
- Removing duplicate and incomplete email records
- Eliminating irrelevant symbols, numbers, and formatting noise
- Standardizing text format for consistency

This step was essential to ensure high-quality input data, as machine learning models are highly sensitive to noisy or inconsistent data. Completing this phase establishes a strong foundation for the entire project.

2. Feature Extraction – 100% Completed

Feature extraction has been fully implemented to convert raw email text into machine-readable numerical representations. AI and machine learning–based feature extraction techniques such as Bag of Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF) have been successfully applied.

These techniques help the system:

- Identify important and discriminative words
- Reduce the impact of commonly occurring but less meaningful terms
- Improve the learning efficiency of machine learning models

Completion of this phase ensures that the system can effectively represent email content for accurate classification.

3. Model Implementation – 70% Completed

The implementation of machine learning models for spam classification is currently 70% complete. Several AI/ML classifiers such as Naïve Bayes, Logistic Regression, Support Vector Machines, and Random Forests have already been implemented and trained on the processed dataset.

However, some work is still in progress, including:

- Fine-tuning model parameters
- Implementing ensemble learning techniques
- Enhancing model robustness for unseen data
- Despite being partially complete, the implemented models already demonstrate promising classification performance.

4. Model Evaluation and Optimization – 60% Completed

The evaluation and optimization phase is currently 60% complete. Initial performance evaluation has been conducted using standard machine learning metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

Ongoing optimization tasks include:

- Reducing false positive rates
- Improving generalization across different datasets
- Comparing individual models with ensemble approaches
- Further tuning and validation are required to finalize the most optimal model configuration.

5. Integration and Testing – 60% Completed

System integration and testing are currently 60% complete. The integration process involves combining data preprocessing, feature extraction, and trained machine learning models into a single unified workflow.

Testing activities include:

- Functional testing of classification outputs
- Validation of model predictions
- Handling edge cases such as empty or noisy email inputs
- Additional testing is required to ensure stability and scalability of the system under real-world conditions.

6. Overall Project Completion Status – 75%

Based on the progress of individual development stages, the overall project completion status is approximately 75%. Core functionalities such as data preparation and feature extraction are fully completed, while advanced model optimization and system integration are in progress.

The project is well on track for successful completion, with remaining work focused on enhancing model performance, completing system integration, and preparing the final deployment and documentation.

7. FUTURE SCOPE

Although the proposed spam detection system delivers high accuracy and reliable performance, there is significant scope for further enhancement and real-world deployment. Some potential future improvements are discussed below:

1. Real-Time Email Filtering

The current system can be extended to work in real-time environments, where emails are classified instantly as they arrive. This would allow direct integration with live email servers and prevent spam emails from ever reaching the user's inbox.

2. Online Learning and Adaptive Models

Future versions of the system can incorporate online or incremental learning, enabling the model to continuously learn from new spam patterns. This would help the system remain effective against newly emerging spam techniques without the need for complete retraining.

3. Phishing and Malware Detection

The system can be enhanced to classify emails into multiple categories such as:

- Phishing emails
- Malware-infected emails
- Promotional spam
- Legitimate emails

This multi-class classification approach would significantly improve cybersecurity.

4. Deep Learning Integration

Advanced deep learning models such as LSTM, GRU, and Transformer-based models (e.g., BERT) can be implemented to better understand contextual relationships between words. These models can further improve detection accuracy, especially for complex and disguised spam emails.

5. Large-Scale Enterprise Deployment

The system can be scaled for enterprise-level usage, where thousands of emails are processed daily. Cloud-based deployment using distributed computing frameworks would allow high-speed processing and better scalability.

6. Multilingual Spam Detection

Currently, the system focuses on English emails. Future enhancements can include multilingual spam detection, allowing classification of spam emails written in different languages, which is especially useful for global email services.

7. User Feedback-Based Learning

Incorporating a user feedback mechanism would allow users to manually mark misclassified emails. This feedback can be used to retrain the model, further reducing false positives and improving personalization.

8. Integration with Other Platforms

The spam detection framework can be extended beyond emails to detect spam in:

- SMS messages
- Social media comments
- Online forums and chat applications

This would make the system a generalized text-based spam detection solution.